# EXPERIMENT 5

## Support Vector Machine (SVM) with Hyperparameter Tuning

**Aim:**
To implement **Support Vector Machine (SVM)** for classification on the Titanic dataset and improve model performance using **Hyperparameter Tuning (GridSearchCV)**.

**Theory:**
Support Vector Machine (SVM) is a supervised machine learning algorithm used for classification and regression problems.

In classification, SVM works by finding an **optimal hyperplane** that separates data points of different classes with the maximum margin.

The main idea of SVM is:
- Find the best boundary that separates classes.
- Maximize the distance (margin) between the nearest data points of both classes.
- These nearest points are called **Support Vectors**.

For non-linear data, SVM uses a **Kernel Trick** such as:
- Linear
- RBF (Radial Basis Function)
- Polynomial

Important Hyperparameters:
1. **C (Regularization Parameter)**
   Controls trade-off between margin size and classification error.

2. **Kernel**
   Defines type of decision boundary.
3. **Gamma**
   Controls how far the influence of a single training example reaches.

Hyperparameter tuning helps in selecting the best combination of these parameters to improve model performance.

**<u>Real-Life Example</u>**
Imagine you are separating:
- Apples
- Oranges

Based on:
- Weight
- Color
- Size

Now imagine drawing a line on a table that separates apples on one side and

oranges on the other. SVM tries to:

- Draw the best possible line
- Keep maximum distance from both fruits
- Make sure fruits are correctly separated

If fruits are mixed in a curved pattern, SVM uses a curved boundary (like RBF kernel). Hyperparameter tuning is like:Trying different types of lines and settings to find the best separation.

**Choosing Dataset:**

| PassengerId<br>Passenger ID | Survived<br>Weather Survived or not: 0 = No, 1 = Yes | Pclass<br>Ticket class: 1 = 1st, 2 = 2nd, 3 = 3rd | Name<br>Name of the Passenger | Sex<br>Gender | Age<br>Age in Years | SibSp<br>No. of siblings aboard the Tita |
|---|---|---|---|---|---|---|
| 1 891 | 0 1 | 1 3 | 891 unique values | male 65%<br>female 35% | 0.42 80 | 0 |
| 1 | 0 | 3 | Braund, Mr. Owen Harris | male | 22 | 1 |
| 2 | 1 | 1 | Cumings, Mrs. John Bradley (Florence Briggs Thayer) | female | 38 | 1 |
| 3 | 1 | 3 | Heikkinen, Miss. Laina | female | 26 | 0 |
| 4 | 1 | 1 | Futrelle, Mrs. Jacques Heath (Lily May Peel) | female | 35 | 1 |
| 5 | 0 | 3 | Allen, Mr. William Henry | male | 35 | 0 |

We choose the titanic Dataset for this experiment.
The sinking of the Titanic is one of the most infamous shipwrecks in history.

While there was some element of luck involved in surviving, it seems some groups of people were more likely to survive than others.

In this challenge, we build a predictive model that answers the question: “what sorts of people were more likely to survive?” using passenger data (ie name, age, gender, socio-economic class, etc).

**Step 1: Data Collection**

Download Titanic dataset from Kaggle.
Use train.csv.

**Step 2: Data Preprocessing**

***Handle Missing Values:***
Age filled using median.
Embarked filled using mode.
Remove Unnecessary Columns:
PassengerId
Name
Ticket
Cabin
Encode Categorical Data:
Sex converted to numeric (male=0, female=1)
Embarked converted using one-hot encoding

Separate:
Features (X)
Target (Survived)

**Step 3: Train-Test Split**

80% for training
20% for testing
Ensures unbiased evaluation

**Step 4: Feature Scaling**

StandardScaler used
Important because SVM is sensitive to feature magnitudes

**Step 5: Model Training (Baseline SVM)**

Train SVM using default parameters
Predict test data
Calculate baseline accuracy

**Step 6: Hyperparameter Tuning**

Use GridSearchCV
Try different combinations of:
C
Kernel
Gamma
5-fold cross-validation used
Select best parameter combination

**Step 7: Model Evaluation**
Evaluate using:
Accuracy Score
Confusion Matrix

Precision
Recall
F1-Score

**CODE :**

```
# ==============================================

# SVM Classification with Hyperparameter Tuning # Dataset: Titanic (Kaggle)
# ==============================================
# Step 1: Import Libraries import numpy as np
import pandas as pd

import matplotlib.pyplot as plt import seaborn as sns


from sklearn.model_selection import train_test_split, GridSearchCV from
sklearn.preprocessing import StandardScaler
from sklearn.svm import SVC
```

```
from sklearn.metrics import accuracy_score, confusion_matrix,
classification_report


# Step 2: Load Dataset (Correct Path)

data = pd.read_csv('sample_data/train.csv')


# Step 3: Basic Data Cleaning


# Fill missing Age with median

data['Age'] = data['Age'].fillna(data['Age'].median())


# Fill missing Embarked with mode

data['Embarked'] = data['Embarked'].fillna(data['Embarked'].mode()[0])


# Drop unnecessary columns

data = data.drop(['PassengerId', 'Name', 'Ticket', 'Cabin'], axis=1)


# Convert categorical variables

data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})


# One-hot encoding for Embarked

data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
```

```
# Step 4: Define Features and Target

X = data.drop('Survived', axis=1) y = data['Survived']


# Step 5: Train-Test Split

X_train, X_test, y_train, y_test = train_test_split( X, y, test_size=0.2,
random_state=42
)



# Step 6: Feature Scaling (Very Important for SVM) scaler = StandardScaler()
X_train = scaler.fit_transform(X_train) X_test = scaler.transform(X_test)


# Step 7: Baseline SVM Model svm_model = SVC() svm_model.fit(X_train,
y_train)


y_pred = svm_model.predict(X_test)

print("Baseline Accuracy:", accuracy_score(y_test, y_pred))



# Step 8: Hyperparameter Tuning using GridSearchCV param_grid = {
'C': [0.1, 1, 10, 100],

'gamma': ['scale', 'auto'],

'kernel': ['linear', 'rbf']

}

grid = GridSearchCV(SVC(), param_grid, cv=5, verbose=1) grid.fit(X_train,
y_train)


print("\nBest Parameters Found:", grid.best_params_)


# Step 9: Evaluate Tuned Model best_model = grid.best_estimator_ y_pred_best =
```

```
best_model.predict(X_test)

print("\nTuned Model Accuracy:", accuracy_score(y_test, y_pred_best))

print("\nConfusion Matrix:\n", confusion_matrix(y_test, y_pred_best))

print("\nClassification Report:\n", classification_report(y_test, y_pred_best))
```

**OUTPUT :**

```
Baseline Accuracy: 0.8212290502793296
Fitting 5 folds for each of 16 candidates, totalling 80 fits

Best Parameters Found: {'C': 1, 'gamma': 'scale', 'kernel': 'rbf'}

Tuned Model Accuracy: 0.8212290502793296

Confusion Matrix:
 [[95 10]
 [22 52]]

Classification Report:
              precision    recall  f1-score   support

           0       0.81      0.90      0.86       105
           1       0.84      0.70      0.76        74

    accuracy                           0.82       179
   macro avg       0.83      0.80      0.81       179
weighted avg       0.82      0.82      0.82       179
```

**Results Summary:**
Baseline Accuracy: 82.12%
Tuned Accuracy: 82.12%
Best Parameters:
C = 1
Kernel = RBF
Gamma = scale

The tuning confirmed that default parameters were already optimal.

**CONCLUSION:**

The Support Vector Machine (SVM) classifier was successfully implemented on the Titanic dataset. Data preprocessing steps such as handling missing values, encoding categorical variables, and feature scaling were performed to prepare the dataset for modeling.
Hyperparameter tuning using GridSearchCV was applied to find the best combination of parameters. The optimized SVM model achieved an accuracy of 82.12%.
The experiment demonstrates that SVM is an effective classification algorithm and hyperparameter tuning helps in selecting the best configuration for improved performance.

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